Two junior data engineers are authoring separate parts of a single data pipeline notebook. They are working on separate Git branches so they can pair program on the same notebook simultaneously. A senior data engineer experienced in Databricks suggests there is a better alternative for this type of collaboration.
Which of the following supports the senior data engineer's claim?

A. Databricks Notebooks support automatic change-tracking and versioning

B. Databricks Notebooks support real-time coauthoring on a single notebook

C. Databricks Notebooks support commenting and notification comments

D. Databricks Notebooks support the use of multiple languages in the same notebook

E. Databricks Notebooks support the creation of interactive data visualizations

A data engineer and data analyst are working together on a data pipeline. The data engineer is working on the raw, bronze, and silver layers of the pipeline using Python, and the data analyst is working on the gold layer of the pipeline using SQL. The raw source of the pipeline is a streaming input. They now want to migrate their pipeline to use Delta Live Tables. Which of the following changes will need to be made to the pipeline when migrating to Delta Live Tables?

A. None of these changes will need to be made

B. The pipeline will need to stop using the medallion-based multi-hop architecture

C. The pipeline will need to be written entirely in SQL

D. The pipeline will need to use a batch source in place of a streaming source

E. The pipeline will need to be written entirely in Python

Given the following two tables, which JOIN operation should be used in the blank to ensure the query returns all students, including those not enrolled in any course, with their course_id if available?

```
SELECT students.name, students.age, enrollments.course_id
FROM students
_____________ enrollments
ON students.student_id = enrollments.student_id
```

A RIGHT JOIN

B INNER JOIN

C LEFT JOIN

D ANTI JOIN

E CROSS JOIN

A data engineer wants to create a data entity from a couple of tables. The data entity must be used by other data engineers in other sessions. It also must be saved to a physical location. Which of the following data entities should the data engineer create?

A. Table

B. Function

C. View

D. Temporary view

A data analyst has a series of queries in a SQL program. The data analyst wants this program to run every day. They only want the final query in the program to run on Sundays. They ask for help from the data engineering team to complete this task. Which of the following approaches could be used by the data engineering team to complete this task?

A. They could submit a feature request with Databricks to add this functionality.

B. They could wrap the queries using PySpark and use Python's control flow system to determine when to run the final query.

C. They could only run the entire program on Sundays.

D. They could automatically restrict access to the source table in the final query so that it is only accessible on Sundays.

E. They could redesign the data model to separate the data used in the final query into a new table.

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You have a DataFrame `df` with a nested column `address.city`. How would you extract this nested field into a top-level column named `city` using Spark? Provide the code snippet.

A. df.withColumn('city', col('address.city'))

B. df.select(col('address.city').alias('city'))

C. df.withColumn('city', 'address.city')

D. df.select('address.city').alias('city')

Which of the following functions is **not** considered a Higher Order Function in Spark SQL?

A TRANSFORM

B EXIST

C FILTER

D REDUCE

E MAP

Which of the following code blocks will remove the rows where the value in column age is greater than 25 from the existing Delta table my_table and save the updated table?

A SELECT * FROM my_table WHERE age> 25;

B UPDATE my_table WHERE age > 25;

C DELETE FROM my_table WHERE age > 25;

D UPDATE my_table WHERE age <= 25;

E DELETE FROM my_table WHERE age <= 25;

Which of the following describes the type of workloads that are always compatible with Auto Loader?

A. Streaming workloads

B. Machine learning workloads

C. Serverless workloads

D. Batch workloads

E. Dashboard workloads

A data engineer has configured a Structured Streaming job to read from a table, manipulate the data, and then perform a streaming write into a new table.
The code block used by the data engineer is below:

```
(spark.table("sales")
  .withColumn("avg_price", col("sales") / col("units"))
  .writeStream
  .option("checkpointLocation", checkpointPath)
  .outputMode("complete")
  ._____
  .table("new_sales"))
```

If the data engineer only wants the query to execute a single micro-batch to process all of the available data, which of the following lines of code should the data engineer use to fill in the blank?

A. trigger(once=True)

B. trigger(continuous="once")

C. processingTime("once")

D. trigger(processingTime="once")

E. processingTime(1)

Which of the following describes the storage organization of a Delta table?

A. Delta tables are stored in a single file that contains data, history, metadata, and other attributes.

B. Delta tables store their data in a single file and all metadata in a collection of files in a separate location.

C. Delta tables are stored in a collection of files that contain data, history, metadata, and other attributes.

D. Delta tables are stored in a collection of files that contain only the data stored within the table.

E. Delta tables are stored in a single file that contains only the data stored within the table.

Describe a scenario where you would need to share a Databricks notebook with a team member who does not have direct access to your workspace. Explain the steps you would take to securely share the notebook and manage access rights.

A. Send the notebook via email without any access control.

B. Use Databricks workspace sharing features to grant specific access rights to the team member.

C. Print the notebook and physically share it.

D. Share the notebook through an external cloud storage service.

What will be the output of the following Python function when called with `my_func(1, 2)`?

```
def my_func(x, y=1):
    if x < y:
        x = 2 * x
    if x == y:
        x += 1
        y -= 2
    return (x, y)
```

A (2, 1)

B (1, 2)

C (3, 0)

D (1, 1)

E (2, 2)

56. In a Job with multiple tasks, how can 'Task A' share a generated ID with 'Task B' downstream?

A	dbutils.jobs.taskValues.set()
B	Write to a DBFS file
C	Cluster environment variables
D	Notebook Widgets

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You are tasked with identifying best practices for colocating metastores with a Databricks workspace. Describe the benefits of this approach and how it impacts performance and data governance. Additionally, explain how you would ensure that the metastore is properly configured to support multiple catalogs and databases.

A. Colocating metastores improves performance by reducing network latency and enhances data governance by centralizing metadata management.

B. Colocating metastores reduces performance by increasing network latency and weakens data governance by decentralizing metadata management.

C. Colocating metastores has no impact on performance or data governance.

D. Colocating metastores improves performance but weakens data governance by centralizing metadata management.

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A data analyst has created a Delta table `sales` that is used by the entire data analysis team. They want help from the data engineering team to implement a series of tests to ensure the data is clean. However, the data engineering team uses Python for its tests rather than SQL. Which of the following commands could the data engineering team use to access `sales` in PySpark?

A	`SELECT * FROM sales`
B	There is no way to share data between PySpark and SQL.
C	`spark.sql("sales")`
D	`spark.delta.table("sales")`
E	`spark.table("sales")`

A data engineer has created a new database using the following command: CREATE DATABASE IF NOT EXISTS customer360;
In which of the following locations will the customer360 database be located?

A. dbfs:/user/hive/database/customer360

B. dbfs:/user/hive/warehouse

C. dbfs:/user/hive/customer360

D. More information is needed to determine the correct response

E. dbfs:/user/hive/database

In a Databricks environment, you are working with a large dataset that includes a 'transaction_date' column formatted as 'yyyy-MM-dd'. Your task is to analyze transactions by year, month, and day. To achieve this, you need to cast the 'transaction_date' column to a timestamp and then extract the year, month, and day from the resulting timestamp. Considering the need for accuracy and performance in processing large datasets, which of the following Spark SQL queries correctly accomplishes this task? Choose the best option from the four provided.

A. SELECT CAST(transaction_date AS TIMESTAMP), EXTRACT(YEAR FROM transaction_date), EXTRACT(MONTH FROM transaction_date), EXTRACT(DAY FROM transaction_date) FROM dataset

B. SELECT transaction_date, YEAR(transaction_date), MONTH(transaction_date), DAY(transaction_date) FROM dataset

C. SELECT CAST(transaction_date AS TIMESTAMP) as ts, EXTRACT(YEAR FROM ts) as year, EXTRACT(MONTH FROM ts) as month, EXTRACT(DAY FROM ts) as day FROM dataset

D. SELECT FROM_UNIXTIME(CAST(transaction_date AS TIMESTAMP), 'yyyy-MM-dd') as formatted_date, EXTRACT(YEAR FROM formatted_date) as year FROM dataset

A junior data engineer is struggling to recall the logic they modified two weeks ago in a data processing task and wishes to revert to the original code. A senior data engineer suggests a more efficient solution to such problems. Which of the following options best supports the senior engineer's advice?

A. Databricks Notebooks enable real-time collaboration on a single notebook

B. Databricks Notebooks allow the use of multiple programming languages within the same notebook

C. Databricks Notebooks feature automatic change-tracking and versioning capabilities

D. Databricks Notebooks support the creation of interactive data visualizations

E. Databricks Notebooks include commenting and notification features

In Azure Databricks, Unity Catalog is a critical component for data governance, providing a unified metadata layer for managing data assets. Considering a scenario where an organization is migrating to Azure Databricks and needs to implement strict data governance policies to comply with GDPR, identify the different types of securables in Unity Catalog that would be most relevant for setting up these policies. The organization is particularly concerned with ensuring that data access is controlled at various levels, including the physical location of data to comply with data residency requirements. Choose the option that best describes the securables relevant for this scenario. (Choose one correct answer)

A Database, Schema, Table, and Data Location

B Database, Schema, Table, and Data Format

C Database, Schema, Table, and Data Type

D Database, Schema, Table, and Data Source

44. Which command grants a specific user the ability to read data from the 'orders' table?

A GRANT USAGE ON TABLE orders TO 'user'

B GRANT MODIFY ON TABLE orders TO 'user'

C GRANT SELECT ON TABLE orders TO 'user'

D GRANT READ ON TABLE orders TO 'user'

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A GRANT USAGE ON TABLE orders TO 'user'

B GRANT MODIFY ON TABLE orders TO 'user'

C GRANT SELECT ON TABLE orders TO 'user'

D GRANT READ ON TABLE orders TO 'user'

A data engineer is tasked with identifying customers who have not placed any orders on an e-commerce platform. They have two Delta tables: `customers`, containing details of registered customers, and `orders`, containing details of orders placed. Which query correctly identifies these customers?

A. SELECT customer_id FROM customers INTERSECT SELECT order_customer_id FROM orders

B. SELECT customer_id FROM customers UNION SELECT order_customer_id FROM orders

C. SELECT customer_id FROM customers c INNER JOIN orders o ON c.customer_id = o.order_customer_id

D. SELECT customer_id FROM customers UNION ALL SELECT order_customer_id FROM orders

E. SELECT customer_id FROM customers EXCEPT SELECT order_customer_id FROM orders

Which of the following is **not** a feature of Auto Loader in Databricks?

A. Auto Loader can detect schema drifts, notify you when schema changes happen, and rescue data that would have been otherwise ignored or lost.

B. Auto Loader provides a SQL warehouse which is a compute resource that lets you run SQL commands on data objects within Databricks SQL.

C. Auto Loader incrementally and efficiently processes new data files as they arrive in cloud storage. It can be used to process billions of files to migrate or backfill a table.

D. Auto Loader's file notification mode can help reduce your cloud costs further by avoiding directory listing altogether. It can automatically set up file notification services on storage to make file discovery much cheaper.

E. Auto Loader can resume from where it left off by information stored in the checkpoint location and continue to provide exactly-once guarantees when writing data into Delta Lake in case of any failures.

Question 33

A data engineer has three notebooks in an ELT pipeline. The notebooks need to be executed in a specific order for the pipeline to complete successfully. The data engineer would like to use Delta Live Tables to manage this process.

Which of the following steps must the data engineer take as part of implementing this pipeline using Delta Live Tables?

A. They need to create a Delta Live Tables pipeline from the Data page.

B. They need to create a Delta Live Tables pipeline from the Jobs page.

C. They need to create a Delta Live tables pipeline from the Compute page.

D. They need to refactor their notebook to use Python and the dlt library.

E. They need to refactor their notebook to use SQL and `CREATE LIVE TABLE` keyword.

As part of the project to create a comprehensive employee performance tracking system, a data architect has specified the need for a new table designed to store employee ratings over time. The required schema for this table is as follows: (employeeId STRING, startDate DATE, avgRating FLOAT).
Which code block, using a SQL Data Definition Language (DDL) command, would you utilize to create an empty Delta table with the aforementioned schema? The solution should ensure that the table is created even if a table with the same name already exists, without causing errors due to existing table conflicts.

A. CREATE OR REPLACE TABLE table_name (employeeId STRING, startDate DATE, avgRating FLOAT)

B. CREATE OR REPLACE TABLE table_name WITH COLUMNS (employeeId STRING, startDate DATE, avgRating FLOAT) USING DELTA

C. CREATE TABLE IF NOT EXISTS table_name (employeeId STRING, startDate DATE, avgRating FLOAT)

D. CREATE TABLE table_name AS SELECT employeeId STRING, startDate DATE, avgRating FLOAT

A data engineer has joined an existing project and they see the following query in the project repository:

```
CREATE STREAMING LIVE TABLE loyal_customers AS SELECT customer_id -
FROM STREAM(LIVE.customers) WHERE loyalty_level = 'high';
```

Which of the following describes why the STREAM function is included in the query?

A. The STREAM function is not needed and will cause an error.

B. The table being created is a live table.

C. The customers table is a streaming live table.

D. The customers table is a reference to a Structured Streaming query on a PySpark DataFrame.

E. The data in the customers table has been updated since its last run.

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E. The data in the customers table has been updated since its last run.

A data engineer has realized that the data files associated with a Delta table are incredibly small. They want to compact the small files to form larger files to improve performance. Which keyword can be used to compact the small files?

A OPTIMIZE

B VACUUM

C COMPACTIO N

D REPARTITION

Question 43

A data engineering manager has noticed that each of the queries in a Databricks SQL dashboard takes a few minutes to update when they manually click the “Refresh” button. They are curious why this might be occurring, so a team member provides a variety of reasons on why the delay might be occurring. Which of the following reasons fails to explain why the dashboard might be taking a few minutes to update?

A	The SQL endpoint being used by each of the queries might need a few minutes to start up.
B	The queries attached to the dashboard might take a few minutes to run under normal circumstances.
C	The queries attached to the dashboard might first be checking to determine if new data is available.
D	The Job associated with updating the dashboard might be using a non-pooled endpoint.
E	The queries attached to the dashboard might all be connected to their own, unstarted Databricks clusters.

As a data engineer, you're tasked with maintaining only the recent versions of the 'sales' Delta table, specifically those not older than the last 7 days. You've noticed the directory for this table contains numerous data files from historical versions. Which query should you execute to remove files from the 'sales' table that are older than one week?

A VACUUM sales RETAIN 7 DAYS

B VACUUM sales RETAIN 604800 SECONDS

C VACUUM sales RETAIN 168 HOURS

D VACUUM sales RETAIN 10080 MINUTES

E VACUUM sales RETAIN 1 WEEK

Which of the following describes a scenario in which a data team will want to utilize cluster pools?

A. An automated report needs to be refreshed as quickly as possible.

B. An automated report needs to be made reproducible.

C. An automated report needs to be tested to identify errors.

D. An automated report needs to be version-controlled across multiple collaborators.

E. An automated report needs to be runnable by all stakeholders.

To ensure optimal performance and management of large-scale data, it's crucial to understand how Delta tables are structured and organized. Given this, which of the following describes the storage organization of a Delta table?

A. Delta tables are stored in a single file that contains data, history, metadata, and other attributes.

B. Delta tables store their data in a single file and all metadata in a collection of files in a separate location.

C. Delta tables are stored in a collection of files that contain data, history, metadata, and other attributes.

D. Delta tables are stored in a collection of files that contain only the data stored within the table.

E. Delta tables are stored in a single file that contains only the data stored within the table.

Mike is the current owner of the `sales` table. He granted Ross the privilege to create a view named `regional_sales_vw`. After creating the view, Ross granted David the privilege to query the view. However, David encounters an error when attempting to access the view. What steps can be taken to resolve this issue?

A. Ross must transfer ownership of the `sales` table from Mike to himself.

B. David needs to become the owner of the `regional_sales_vw` view to access it.

C. Mike must grant David the `SELECT` privilege on the `sales` table.

D. David must own the `sales` table to access the `regional_sales_vw` view.

E. Ross should grant David the `SELECT` privilege on the `sales` table.

A data engineer observed that some unused data files in a Delta table's directory were not deleted after executing the VACUUM command, despite others being removed. What could be the reason for this partial deletion?

A. The deleted data files were larger than the default size threshold, while the remaining files are smaller and cannot be deleted.

B. The deleted data files were newer than the default retention threshold, while the remaining files are older and cannot be deleted.

C. The deleted data files were older than the default retention threshold, while the remaining files are newer and cannot be deleted.

D. The deleted data files were smaller than the default size threshold, while the remaining files are larger and cannot be deleted.

E. More information is needed to determine the correct answer.

A data engineer needs to determine whether to use the built-in Databricks Notebooks versioning or version their project using Databricks Repos. Which of the following is an advantage of using Databricks Repos over the Databricks Notebooks versioning?

A. Databricks Repos automatically saves development progress

B. Databricks Repos supports the use of multiple branches

C. Databricks Repos allows users to revert to previous versions of a notebook

D. Databricks Repos provides the ability to comment on specific changes

E. Databricks Repos is wholly housed within the Databricks Lakehouse Platform

A data engineer that is new to using Python needs to create a Python function to add two integers together and return the sum. Which of the following code blocks can the data engineer use to complete this task?

A

```
function add_integers(x, y):
    return x + y
```

B

```
function add_integers(x, y):
    x + y
```

C

```
def add_integers(x, y):
    print(x + y)
```

D

```
def add_integers(x, y):
    return x + y
```

E

```
def add_integers(x, y):
    x + Y
```

50. A data engineer wants a fast copy of a table for testing, created in seconds without doubling storage costs. Which command is best?

A SHALLOW CLONE

B DEEP CLONE

C CREATE TABLE AS SELECT

D INSERT INTO

Question 14 A data engineer wants to create a relational object by pulling data from two tables. The relational object must be used by other data engineers in other sessions. In order to save on storage costs, the data engineer wants to avoid copying and storing physical data.
Which of the following relational objects should the data engineer create?

A	View
B	Temporary view
C	Delta Table
D	Database
E	Spark SQL Table

A data engineering team is optimizing performance and notices a query on the `orders` Delta table, which contains a large amount of data, is running too slowly due to many small files. They aim to trigger compaction for the year 2022 data only using the `OPTIMIZE` command. Which command should they use?

A. OPTIMIZE TABLE orders WHEN order_date >= '2022-01-01'

B. OPTIMIZE orders WHERE order_date >= '2022-01-01'

C. OPTIMIZE TABLE orders WHERE order_date >= '2022-01-01'

D. OPTIMIZE orders FILTER BY order_date >= '2022-01-01'

E. OPTIMIZE TABLE orders USING order_date >= '2022-01-01'